• The trick estimation

In general, it is not always easy to prove that limit statements are true using the ε, δ definition.
Sometimes we can prove complicated limits using a trick called estimation.

- **The main idea of estimation**

Estimation is a useful trick which can sometimes help us prove complicated limits.

- It basically means replacing a complicated piece of some expression by something larger, but easier to understand.

Actually, this trick - replacing complicated functions by simpler, larger functions - is useful in many different parts of mathematics, not just calculus.

Example

Use estimation to prove that $\lim_{x \to 2} x^3 = 8$.

Solution.

- Guessing a value for δ.

Let $\varepsilon > 0$ be given. We want to find a number $\delta > 0$ so that

$$0 < |x - 2| < \delta \quad \Longrightarrow \quad |x^3 - 8| < \varepsilon.$$

To connect $|x^3 - 8|$ with $|x - 2|$, we factorize

$$|x^3 - 8| = |x^3 - 2^3| = |(x - 2)(x^2 + 2x + 4)| = |x - 2| \cdot |x^2 + 2x + 4|.$$

Then we want

$$0 < |x - 2| < \delta \quad \Longrightarrow \quad |x - 2| \cdot |x^2 + 2x + 4| < \varepsilon.$$

Question. How can we process from $|x-2| \cdot |x^2+2x+4|$ to $|x-2|$?

Answer. We can try to replace the complicated term $|x^2+2x+4|$ by a constant $C > 0$ so that $|x^2+2x+4| < C$ around the point $x = 2$.

If we can find such a C, then

$$|x^3 - 8| = |x-2| \cdot |x^2+2x+4| < C|x-2|.$$

In this case, we can make $C|x-2| < \varepsilon$ by taking $|x-2| < \dfrac{\varepsilon}{C}$, and so we could choose $\delta = \dfrac{\varepsilon}{C}$.

However, $|x^2+2x+4|$ has no upper bound in its domain $\mathbb{R}$. We can only find such a number C if we restrict x to be in some interval centered at 2.

In fact, since we are interested only in values of x that are close to 2, it is reasonable to assume that x is within a distance 1 from 2, that is $|x-2|<1$.

Then we have $1<x<3$, which gives $7<x^2+2x+4<19$. Thus $|x^2+2x+4|<19$, and hence $C=19$ is a suitable choice for the constant.

But now there are two restrictions on $|x-2|$, namely

$$|x-2|<1 \quad \text{and} \quad |x-2|<\frac{\varepsilon}{C}=\frac{\varepsilon}{19}.$$

To make sure that both of these inequalities are satisfied, we take δ to be the smaller of the two numbers 1 and $\frac{\varepsilon}{19}$. The notation for this is $\delta=\min\left\{1,\frac{\varepsilon}{19}\right\}$.

- Showing that this δ works.

Given $\varepsilon > 0$, let $\delta = \min\left\{1, \frac{\varepsilon}{19}\right\}$. If $0 < |x-2| < \delta$, then

$$|x-2| < 1 \implies 1 < x < 3 \implies |x^2+2x+4| < 19.$$

We also have $|x-2| < \frac{\varepsilon}{19}$, and so

$$|x^3-8| = |x-2| \cdot |x^2+2x+4| < \frac{\varepsilon}{19} \cdot 19 = \varepsilon.$$

Therefore, by the definition of limit,

$$\lim_{x \to 2} x^3 = 8.$$

Comment. It is not always easy to prove that limit statements are true using the ε, δ definition.

Fortunately, this is unnecessary, the limits of complicated functions can be found rigorously from the Limit Laws without resorting to the definition directly.

Theorem

Let L, M, a, and k be real numbers and $\lim_{x\to a} f(x) = L$, $\lim_{x\to a} g(x) = M$. Then the following rules are true.

- *Sum/Difference rules:* $\lim_{x\to a} \big(f(x) \pm g(x)\big) = L \pm M$
- *Constant multiple rule:* $\lim_{x\to a} \big(k \cdot f(x)\big) = k \cdot L$
- *Product rule:* $\lim_{x\to a} \big(f(x) \cdot g(x)\big) = L \cdot M$
- *Quotient rule:* $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{L}{M}$, *provided that* $M \neq 0$
- *Power rule:* $\lim_{x\to a} [f(x)]^n = L^n$, *$n$ is a positive integer*
- *Root rule:* $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{L}$, *$n$ is a positive integer*
  (If n is even, we assume that $f(x) \geq 0$ for all x in an interval containing a.)

• The triangle inequality

In many cases, for simplifying complicated expressions and proving limits, we also us the following triangle inequality.

Triangle Inequality

For any numbers a, b, we always have

$$|a+b| \leq |a| + |b|.$$

The triangle inequality is useful in mathematical analysis for determining the best upper estimate on the sum of two or more numbers.

Here we can use the triangle inequality to prove, for example, the Sum rule above.

• Direct Substitution Property

We can check that if f is a polynomial or a rational function and a is in the domain of f, then

$$\lim_{x \to a} f(x) = f(a).$$

Note, however, that not all limits can be evaluated by direct substitution.

Example

Evaluate the limit $\lim_{x\to 0} \dfrac{\sqrt{x^2+1}-1}{x^2}$.

Solution. We cannot apply the Quotient rule immediately, since the limit of the denominator is 0. But we can rationalize the numerator as follows.

$$\begin{aligned}
\lim_{x\to 0} \frac{\sqrt{x^2+1}-1}{x^2} &= \lim_{x\to 0} \frac{\sqrt{x^2+1}-1}{x^2} \cdot \frac{\sqrt{x^2+1}+1}{\sqrt{x^2+1}+1} \\
&= \lim_{x\to 0} \frac{(x^2+1)-1}{x^2(\sqrt{x^2+1}+1)} = \lim_{x\to 0} \frac{x^2}{x^2(\sqrt{x^2+1}+1)} \\
&= \lim_{x\to 0} \frac{1}{\sqrt{x^2+1}+1} = \frac{1}{\lim\limits_{x\to 0}\left(\sqrt{x^2+1}+1\right)} = \frac{1}{2}.
\end{aligned}$$

• The one-sided limits theorem

Some limits are best calculated by first finding the left- and right-hand limits. The following result gives an alternative method to prove the existence of a (two-sided) limit via one-sided limits exist.

Theorem (The one-sided limits)

Let $f : D \to \mathbb{R}$ be a function. Then the limit $\lim_{x\to a} f(x)$ exists if and only if the two one-sided limits both exist and are equal. In this case

$$\lim_{x\to a} f(x) = \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x).$$

Note. When computing one-sided limits, we can use the fact that the Limit Laws also hold for this type of limits.

• Limit does not exist

Now we come back to the question: How to prove that a limit does not exist?

The one-sided limits theorem above gives us the following good way to prove $\lim_{x \to a} f(x)$ does not exit.

Theorem (Limit does not exist)

If either of the one-side limits $\lim_{x \to a^-} f(x)$, $\lim_{x \to a^+} f(x)$ does not exits, or both one-side limits exit, but they are not equal, then so does the limit $\lim_{x \to a} f(x)$.

Example

Prove that the following limit does not exist.

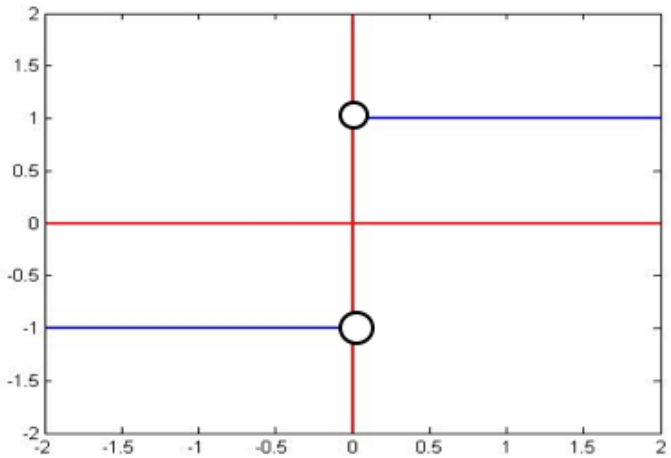
$$\lim_{x \to 0} \frac{|x|}{x}.$$

Solution. We compute the one-side limits

$$\lim_{x \to 0^+} \frac{|x|}{x} = \lim_{x \to 0^+} \frac{x}{x} = \lim_{x \to 0^+} 1 = 1$$

$$\lim_{x \to 0^-} \frac{|x|}{x} = \lim_{x \to 0^-} \frac{-x}{x} = \lim_{x \to 0^-} (-1) = -1.$$

Since the right- and left-hand limits exist, but they are different, we conclude that $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.



The next two theorems give two additional properties of limits.

Theorem

Suppose

- *There exist the limits* $\lim_{x \to a} f(x) = A$ & $\lim_{x \to a} g(x) = B$
- $f(x) \leq g(x)$ *for all* x *in some open interval containing* a*, except possibly at* $x = a$ *itself.*

Then

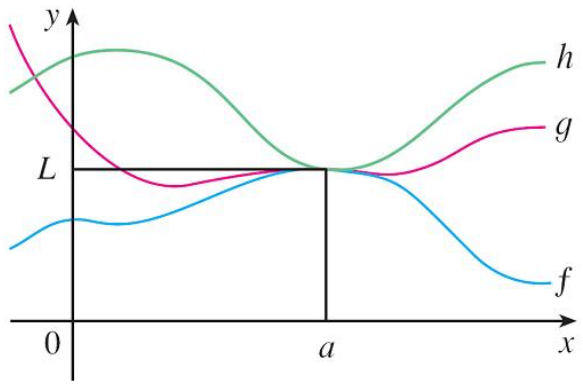
$$\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x), \quad \text{that is} \quad A \leq B.$$

Theorem (The Squeeze Theorem)

Suppose $f(x) \leq g(x) \leq h(x)$ *for all* x *in some open interval containing* a*, except possibly at* $x = a$ *itself. Suppose that*

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L.$$

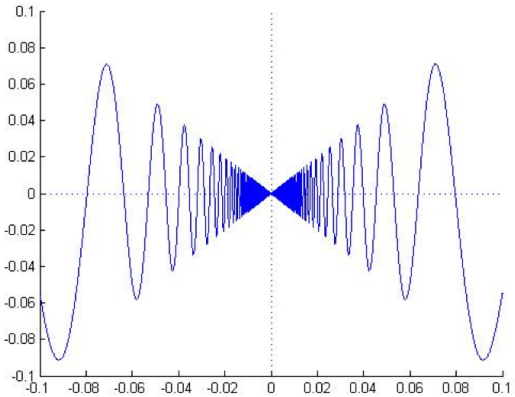
Then there exists $\lim_{x \to a} g(x) = L$.



This theorem says that if $g(x)$ is squeezed between $f(x)$ and $h(x)$ near a, and if f and h have the same limit L at a, then g is forced to have the same limit L at a.

Example

Evaluate the value of the limit $\lim_{x \to 0} x \sin \frac{1}{x}$.



This plot leads us to expect that $\lim_{x \to 0} x \sin \frac{1}{x} = 0$.

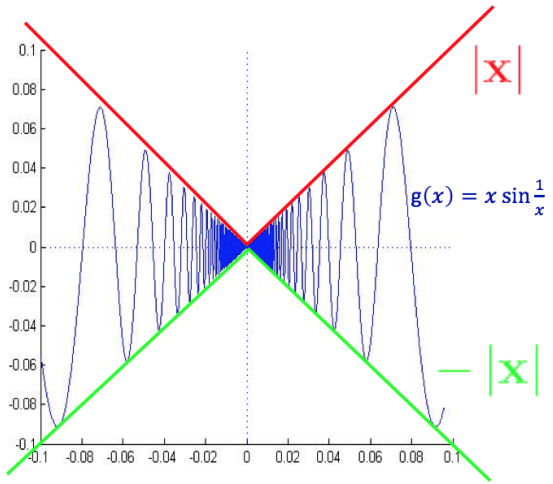
But this limit is not calculated by a combination of the various limit laws. Instead we need the Squeeze Theorem to determine the limit.

Solution. Usually when we use the Squeeze Theorem we have two questions to answer:

(1) What should we set $f(x)$ to be?

(2) What should we set $h(x)$ to be?

Answers. $f(x) = -|x|$ and $h(x) = |x|$. Also we denote $g(x) = x \sin \frac{1}{x}$. These functions work well for the Squeeze Theorem.



◇ **The following section "Extensions of the limit notion" is for a self-study (!).**

1. **Infinite limits**:

$$\lim_{x \to a} f(x) = \pm\infty.$$

2. **Finite limits at infinity**:

$$\lim_{x \to \pm\infty} f(x) = L.$$

3. **Infinite limits at infinity**:

$$\lim_{x \to \pm\infty} f(x) = \pm\infty.$$

• Extensions of the limit notion

Convention. The symbol for infinity ∞ does not represent a real number. We use ∞ to describe the behavior of a function when the values in its domain or range outgrow all finite bounds.

We already studied $\lim_{x \to a} f(x) = L$, where both a and L are real numbers.

We introduce new notions: for real numbers m and $r > 0$, we call

$$V(m, r) := (m - r, m + r),$$

and

$$V_0(m, r) := (m - r, m) \cup (m, m + r) = V(m, r) \setminus \{m\},$$

an r-neighborhood of m and an r-delete neighborhood of m.

Then the ε, δ definition of the limit

$$\lim_{x \to a} f(x) = L$$

can be rephrased as follows:

for any ε-neighborhood of L, there exists a corresponding δ-deleted neighborhood of a, such that $f(x) \in V(L, \varepsilon)$ for all $x \in V_0(a, \delta)$.

Using the idea above, we can extend this notion to the cases when either of a and L is infinity, or both are infinity.

To do that, we define a neighborhood of infinity as follows:

- A neighborhood of $+\infty$ (sometimes also use ∞) is a set $\{x : x > M\}$, where $M > 0$ is an any (large) real number.
- A neighborhood of $-\infty$ is a set $\{x : x < N\}$, where $N < 0$ is an any (small) real number.
- A neighborhood of $\pm\infty$ (sometimes use also ∞) is a set $\{x : x < -E\} \bigcup \{x : x > E\}$, where $E > 0$ is an any (large) real number.

So we consider the following cases:

1. (Infinite limits)

$$\lim_{x \to a} f(x) = \pm\infty.$$

2. (Finite limits at infinity)

$$\lim_{x \to \pm\infty} f(x) = L.$$

3. (Infinite limits at infinity)

$$\lim_{x \to \pm\infty} f(x) = \pm\infty.$$

Definition of infinite limit. Let $f(x)$ be a function defined on some open interval that contains a, except possibly at a itself. We say that

- $\lim_{x \to a} f(x) = \infty$ (or also $+\infty$), if for every positive number M there is a positive number δ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad f(x) > M.$$

- $\lim_{x \to a} f(x) = -\infty$, if for every negative number N there is a positive number δ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad f(x) < N.$$

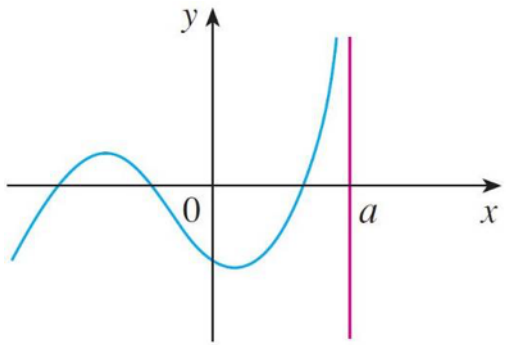
• One-sided infinite limits

Similar definitions can be given for the one-sided infinite limits

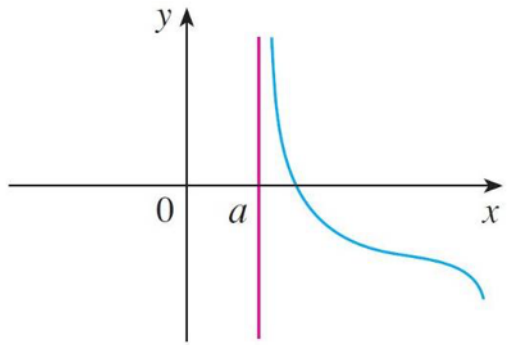
$$\lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty.$$

$$\lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty,$$

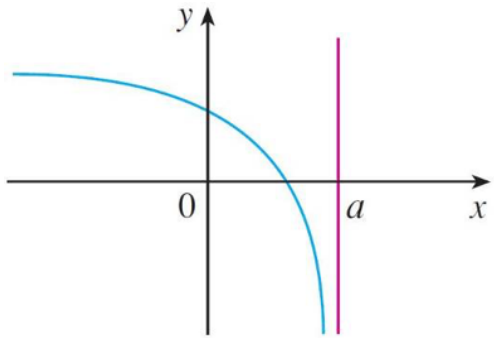
remembering that $x \to a^-$ means that we consider only values of $x < a$, and similarly $x \to a^+$ means that we consider only $x > a$.



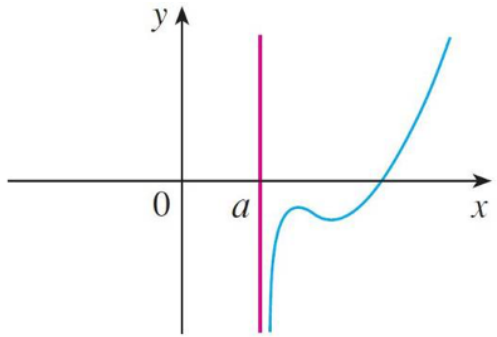
(a) $\lim_{x \to a^-} f(x) = \infty$



(b) $\lim_{x \to a^+} f(x) = \infty$



(c) $\lim_{x \to a^-} f(x) = -\infty$



(d) $\lim_{x \to a^+} f(x) = -\infty$

Definition of finite limit at infinity. We say that

- $f(x)$ has the limit L as x tends to infinity and write $\lim_{x\to\infty} f(x) = L$, if for every number $\varepsilon > 0$, there exists a corresponding number $M > 0$, such that for all x in the domain of f

$$x > M \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

- $f(x)$ has the limit L as x tends to negative infinity and write $\lim_{x\to-\infty} f(x) = L$, if for every number $\varepsilon > 0$, there exists a corresponding number $N < 0$, such that for all x in the domain of f

$$x < N \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Limits at infinity have properties similar to those of finite limits.

Theorem

All the Limit Laws (for finite limits) are true when we replace $\lim_{x\to a}$ *by* $\lim_{x\to\infty}$ *or* $\lim_{x\to-\infty}$ *.*
That is, the variable x may approach a finite number or $\pm\infty$

Example

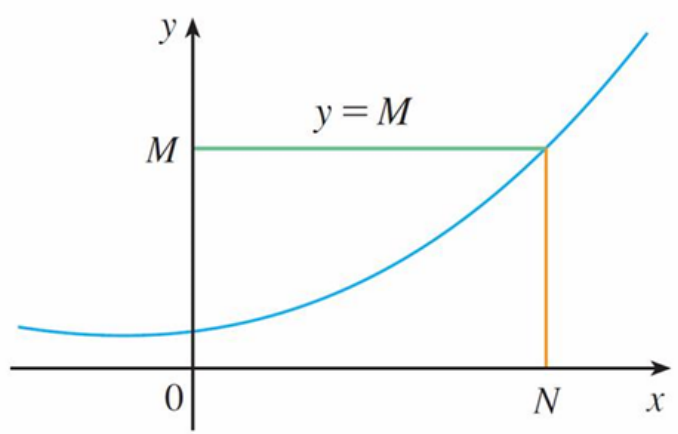
$$\begin{aligned}
\lim_{x\to\pm\infty}\left(1+\frac{2\pi\sqrt{5}}{x^2}\right) &= \lim_{x\to\pm\infty}1+\lim_{x\to\pm\infty}\left(2\pi\sqrt{5}\cdot\frac{1}{x}\cdot\frac{1}{x}\right)\\
&= \lim_{x\to\pm\infty}1+\lim_{x\to\pm\infty}2\pi\sqrt{5}\cdot\lim_{x\to\pm\infty}\frac{1}{x}\cdot\lim_{x\to\pm\infty}\frac{1}{x}\\
&= 1+2\pi\sqrt{5}\cdot 0\cdot 0 = 1+0 = 1.
\end{aligned}$$

Definition of infinite limit at infinity. We say that

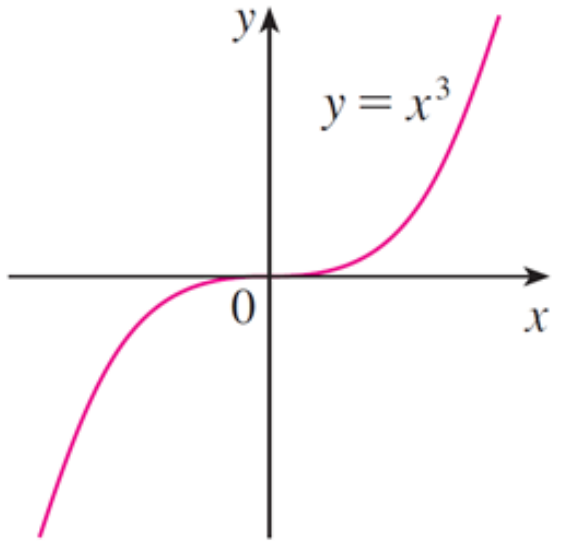
- Let f be a function defined on some interval (a, ∞). We write $\lim_{x \to \infty} f(x) = \infty$, if for every number $M > 0$, there exists a number $N > 0$, such that

$$x > N \quad \Longrightarrow \quad f(x) > M.$$

- Similar definitions apply when the symbol ∞ is replaced by $-\infty$.



$$\lim_{x \to \infty} f(x) = \infty$$



$$\lim_{x \to \infty} x^3 = \infty, \quad \lim_{x \to -\infty} x^3 = -\infty$$

Applications of limits in real life

The concept of limits in mathematics has **numerous applications** in real life across various fields.

There are some examples highlighting a few of the many applications of limits in real-life scenarios.

Limits provide a powerful mathematical framework for understanding change, rates, and continuity ..., which are essential concepts in various scientific, technological, and economic disciplines.

• **Physics**: Limits are used extensively in physics to describe the behavior of physical systems. For instance, when studying the motion of an object, limits help determine instantaneous velocity and acceleration at a specific point in time.

• **Engineering**: Engineers often use limits to analyze and design structures, machines, and systems. They rely on limits to calculate stress and strain in materials, determine the stability of structures, and optimize performance.

• **Computer Science**: Limits are applied in computer science algorithms and computational modeling. They play a crucial role in analyzing the efficiency and complexity of algorithms and determining the behavior of programs.

• **Data Analysis**: When dealing with large datasets, limits are employed to analyze trends, patterns, and anomalies. They allow statisticians and data scientists to make predictions, estimate parameters, and draw conclusions from limited or incomplete information.

• **Probability and Statistics**: Limits are fundamental to probability and statistics. They are used to define concepts like expected value, probability density functions, and cumulative distribution functions, enabling the analysis of random events and data.

• **Medicine**: Limits are utilized in medical research and practice. For instance, in pharmacokinetics, limits help determine the rate at which a drug is absorbed, distributed, metabolized, and excreted in the body.

• **Finance**: Limits are used in finance to study concepts such as compound interest and continuous growth models. They help evaluate investment strategies, predict future returns, and assess the risk associated with financial instruments.

• **Economics**: In economics, limits are used to study rates of change and marginal analysis. They help economists understand how quantities such as demand, supply, and production change with respect to various factors like price, time, and inputs.

Outline

1 Functions

2 Limits

3 Continuity

Summary

In this section, we study the following topics:

- Continuity of a function at a point; continuity test
- Continuous functions and their properties
- Important classes of continuous functions
- The application: Intermediate Value Theorem for continuous functions

Observation

We noticed previously that the limit of a function as x tends to a can often be found simply by calculating the value of the function at a.

We already studied the "Direct Substitution Property": If f is a *polynomial* or a *rational function* and a is in the domain of f, then

$$\lim_{x \to a} f(x) = f(a).$$

For example, if

$$f(x) = \frac{x+1}{x^2-2},$$

then

$$\lim_{x \to 1} f(x) = \lim_{x \to 1} \frac{x+1}{x^2-2} = f(1) = -2.$$

• Continuity

It is true that for *many functions* $f(x)$ we can obtain the limit $\lim_{x\to a} f(x)$ just by evaluating the function at the point a we are taking the limit.

These are actually very special types of functions: the continuous functions.

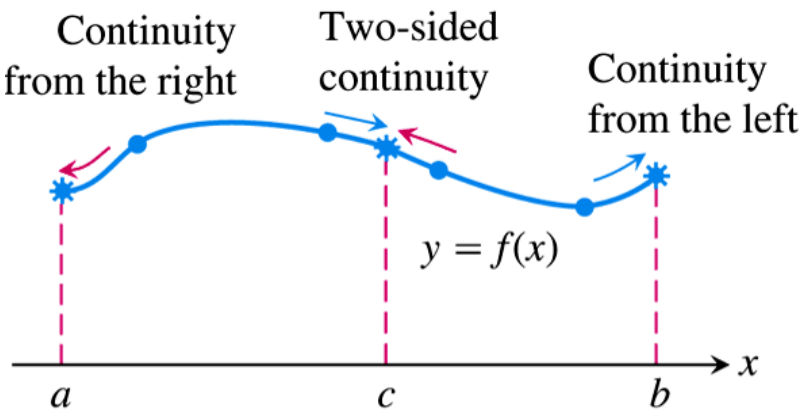
Definition. Let f be a function and let $a \in \mathbb{R}$. This function is said to be

- continuous at a, if $\lim_{x\to a} f(x) = f(a)$.
- right-continuous (or continuous from the right) at a, if $\lim_{x\to a^+} f(x) = f(a)$.
- left-continuous (or continuous from the left) at a, if $\lim_{x\to a^-} f(x) = f(a)$.

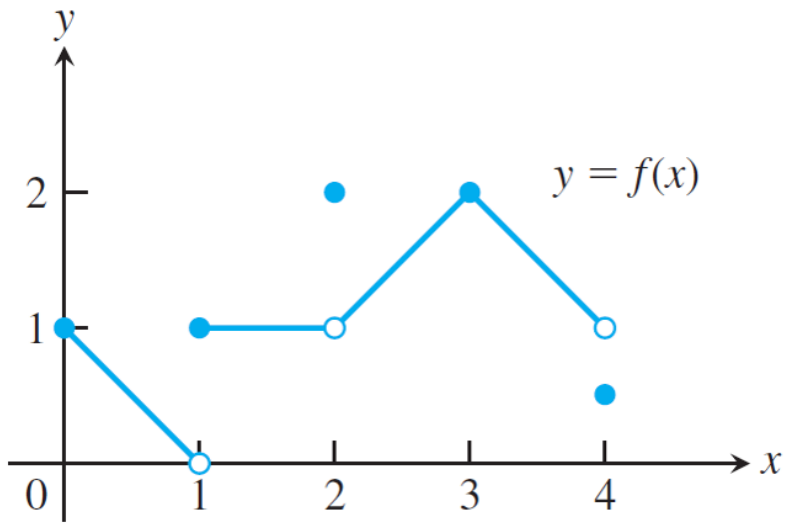
Note.

- The equation in the definition is sometimes called the direct substitution property.
- The intuitive idea of a continuous function is that the graph can be drawn without lifting the pen off the paper. Or, *a small change in x away from $x = a$ only leads to a small change in $f(x)$*.

Below is a continuity at points $x = a$, $x = b$, and $x = c$

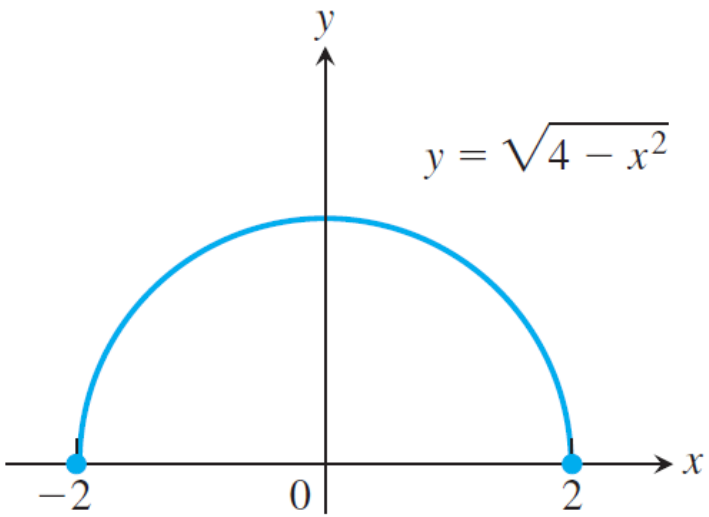


In the figure below, the function is not continuous at points $x = 1$, $x = 2$, and $x = 4$.



Example

The function $f(x) = \sqrt{4 - x^2}$ is continuous over its domain $[-2, 2]$. It is right-continuous at $x = -2$, and left-continuous at $x = 2$.



We summary continuity at an interior point in the form of a test, which can be easily obtained from the definition.

Continuity Test

A function $f(x)$ is continuous at the point $x = a$ if the following three conditions are true:

(i) $f(a)$ is defined (*that is, a is in the domain of f*)

(ii) The limit $\lim_{x \to a} f(x)$ exists (*that is, f has a limit as* $x \to a$)

(iii) $\lim_{x \to a} f(x) = f(a)$ (*that is, the limit equals the function value*).

Note. For one-side continuity and continuity at an endpoint of an interval, the limits in parts (ii) and (iii) of the test should be replaced by the appropriate one-side limits.

• Discontinuity

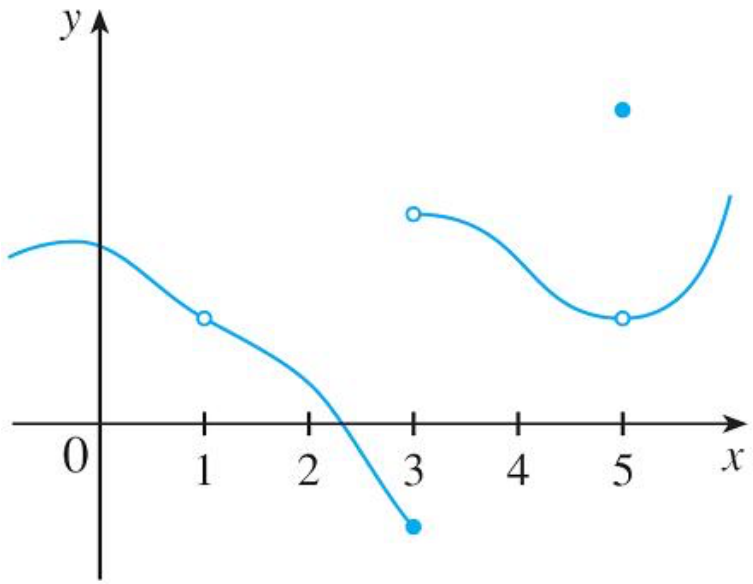
If a function f is defined near a (that is, f is defined on an open interval containing a, except perhaps at a), and f is not continuous at a, then we say that f is discontinuous at a (or f has a discontinuity at a).

Comments. From the Continuity Test, it follows that a function f is NOT continuous at a, if and only if either of the following conditions fail to hold:

- $f(a)$ is not defined.
- The limit $\lim_{x \to a} f(x)$ does not exist.
- $\lim_{x \to a} f(x) \neq f(a)$.

Exercise

Figure below shows the graph of a function f. At which numbers is f discontinuous? Why?



• Classification of discontinuity

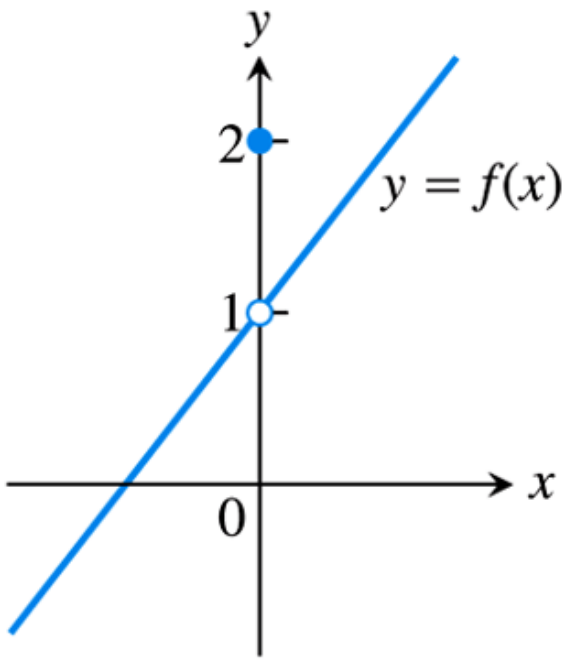
Suppose a function $f(x)$ is defined near a, but it is discontinuous at $x = a$.

- If the limit $\lim_{x \to a} f(x) = L$, but $L \neq f(a)$, then a is called a removable discontinuity.

This discontinuity can be removed to make f continuous at $x = a$, or more precisely, the function

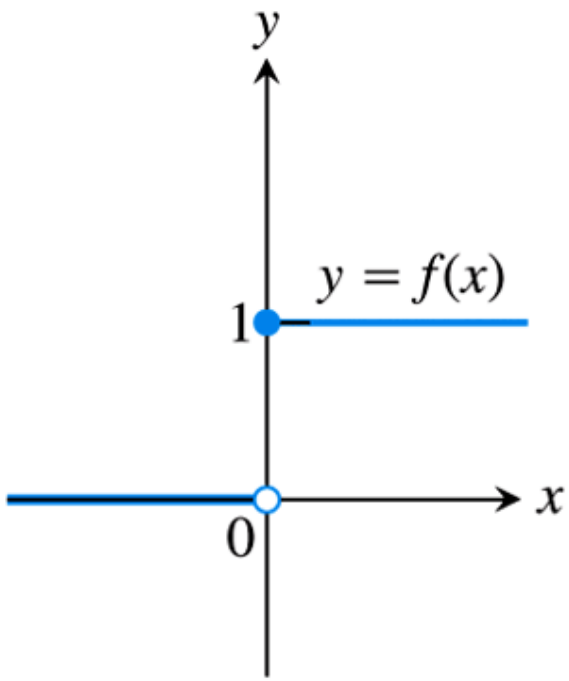
$$g(x) = \begin{cases} f(x) & x \neq a \\ L & x = a \end{cases}$$

is continuous at $x = a$.

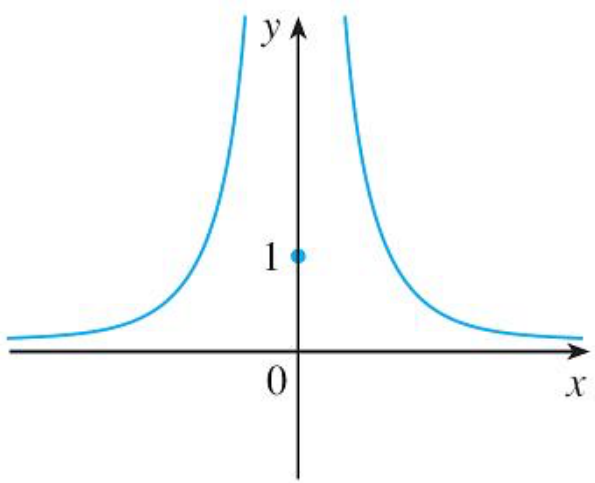

y
2
1
0
x
y = f(x)

- If the two one-sided limits exist and are finite, but are not equal (and hence the limit L does not exist), then a is called a jump discontinuity (or discontinuity of the first kind).

For this type of discontinuity, the function f may have any value at a.



- If one of the two one-sided limits does not exist or is infinite, then a is called an essential discontinuity (or discontinuity of the second kind).



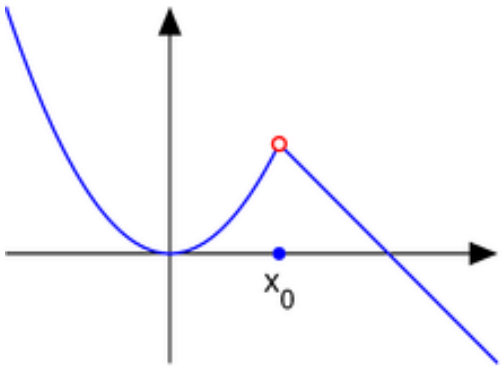
Exercise

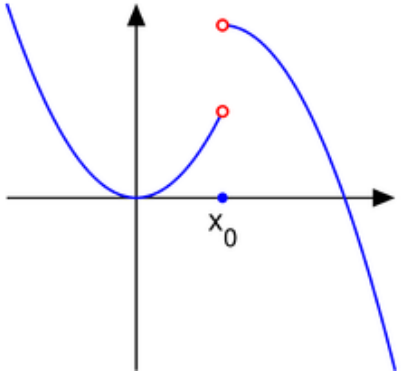
Classify discontinuity of the following functions at $x_0 = 1$.

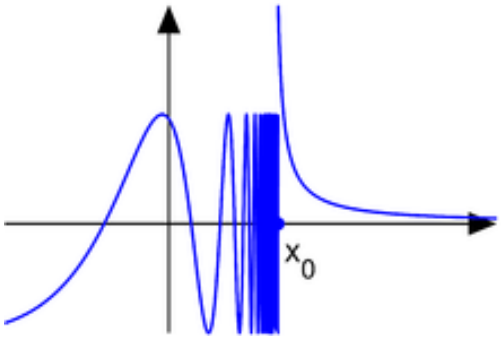
$$(a) \quad f(x) = \begin{cases} x^2, & x < 1 \\ 0, & x = 1 \\ 2 - x, & x > 1 \end{cases} \qquad (b) \quad f(x) = \begin{cases} x^2, & x < 1 \\ 0, & x = 1 \\ 2 - (1 - x)^2, & x > 1 \end{cases}$$

$$(c) \quad f(x) = \begin{cases} \sin \dfrac{5}{x - 1}, & x < 1 \\ 0, & x = 1 \\ \dfrac{1}{x - 1}, & x > 1 \end{cases}$$

The graphs of these functions are shown in the next slide.


x_0


x_0


x_0